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Cost Structure and Channel-Wise Profitability Analysis for Multi-Channel Restaurants

Executive Summary

The analysis clearly indicates a structural imbalance in the company's revenue vs profitability mix. While third-party aggregators—Uber Eats and DoorDash—are driving a significant 61.12% of total revenue, their contribution to overall profitability is disproportionately low at just 5.21%.

Both platforms are operating at extremely thin profit margins (0.84 and 0.90 respectively), which signals that the business is scaling revenue at the cost of profitability. This creates a long-term risk where higher order volumes do not translate into financial sustainability.

In contrast, In-Store and Self-Delivery channels together contribute 94.79% of total profits, making them the true value-generating engines of the business.

Problem Statement

Despite having access to detailed cost and profit data, restaurant operators often lack:

- A side-by-side comparison of channel profitability
- Visibility into commission drag vs delivery cost
- Understanding of which channels erode margins despite high revenue
- Evidence to guide channel prioritization or renegotiation strategies

As a result, many restaurants unintentionally scale low-margin or loss-making channels.

Dataset Description

The dataset contains detailed operational, financial, and channel-level data for multi-channel restaurant businesses, including:

- Restaurant Information: Restaurant ID, name, cuisine type, business segment (e.g., café, QSR), and subregion
- Sales & Order Data: Total monthly orders and channel-wise order distribution across in-store, Uber Eats, DoorDash, and self-delivery
- Revenue Metrics: Channel-wise revenue generated from in-store dining, aggregator platforms, and self-delivery
- Cost Structure Data: Cost of Goods Sold (COGS rate), operating expenses (OPEX rate), commission rates charged by delivery platforms, and delivery cost per order
- Delivery Operations: Delivery radius and total self-delivery logistics cost
- Profitability Metrics: Pre-calculated net profit for each channel after accounting for relevant costs (COGS, OPEX, commissions, and delivery costs)
- Channel Mix Indicators: Percentage share of orders across in-store, aggregator platforms, and self-delivery

Forecasting Methodology (Step-by-Step)

Profit Calculation Validation

- Validate provided net profit formulas
- Normalize profits on a per-order basis to ensure fair comparison
- Identify outliers or extreme margin cases

Channel Margin Analysis

- Compute:
  - Net profit per channel
  - Margin percentage per channel
- Compare absolute profit vs margin efficiency

Cost Component Decomposition

- Break down revenue into:
  - COGS impact
  - OPEX impact
  - Commission drag (Uber Eats, DoorDash)
  - Delivery logistics cost (Self-Delivery)
- Visualize cost waterfalls per channel

Commission & Delivery Cost Impact Analysis

- Assess sensitivity of net profit to:
  - CommissionRate changes
  - DeliveryCostOrder increases
- Identify breakeven thresholds for self-delivery

Cuisine & Segment Profitability Comparison

- Channel-wise margin comparison by:
  - CuisineType
  - Business Segment (Cafe vs QSR)
- Identify margin-resilient vs margin-fragile categories

Profit Volatility & Risk Assessment

- Measure margin variability across restaurants
- Identify channels with unstable profitability
- Flag loss-prone channel strategies

Primary Objectives

- Compare net profitability across In-Store, Uber Eats, DoorDash, and Self-Delivery
- Quantify margin erosion due to commissions and delivery costs
- Identify the most cost-efficient channel per restaurant type

Secondary Objectives

- Support data-driven channel prioritization
- Provide financial evidence for aggregator negotiations
- Highlight self-delivery ROI thresholds

Primary Objectives & Key Findings

1. Net Profitability Comparison Across Channels

A comparative analysis across all four channels—In-Store, Self-Delivery, Uber Eats, and DoorDash—reveals a clear hierarchy in profitability:

- In-Store and Self-Delivery channels are consistently profitable across all segments, showing stable margins and strong cost control.
- Uber Eats and DoorDash, despite high order volumes, fail to convert revenue into meaningful profit, largely due to high commission structures and delivery-related costs.

This reinforces the need to shift strategic focus from revenue-heavy channels to profit-efficient channels.

2. Margin Erosion Due to Cost Structures

- The primary driver of margin erosion is the high commission rates charged by aggregators, particularly in the ≥30% commission bracket, which alone contributes to over 62% of total commission outflow.
- When combined with delivery and operational costs, these commissions significantly compress margins, in some cases pushing them into negative territory, especially in certain restaurant segments.

This is a critical financial inefficiency that needs immediate correction through pricing or negotiation.

3. Identification of Cost-Efficient Channels

From a cost-efficiency perspective:

- In-Store and Self-Delivery emerge as the most efficient and scalable channels, offering high margins and operational control.
- Aggregator platforms (Uber Eats and DoorDash) act more as demand generators rather than profit centers, and should be treated strategically rather than operationally.

Deep-Dive Insights & Segment-Level Analysis

Channel Contribution Dynamics

- 94.79% of total profit is generated by In-Store and Self-Delivery channels
- Aggregators, despite high revenue share, contribute minimally to profit
- This clearly suggests that future growth strategies should prioritize owned channels, while aggregators should be optimized, not expanded blindly.

Segment-Wise Performance Analysis

Full-Service Restaurants

- Negative profit margins observed in:
  - Uber Eats
  - DoorDash
- Other channels remain profitable

Interpretation :

This segment is highly sensitive to commission structures and cost leakage.

Action Required :

- Conduct detailed financial audits
- Identify pricing gaps and cost inefficiencies
- Bring aggregator channels at least to break-even level

Ghost Kitchens

- Strong and consistently positive margins across all channels

Opportunity:

- This is the most scalable and efficient business model

Recommended Actions :

- Expand SKU offerings
- Increase customer acquisition efforts
- Improve In-Store equivalent performance (currently low monthly orders)

Café & QSR Segment

- Profit margins are positive but significantly compressed on aggregator platforms

Challenge:

- High dependency on aggregators but low margin realization

Action Plan :

- Optimize pricing strategies
- Target margin improvement to 10–12%
- Reduce reliance on discount-heavy orders

Uber Eats Revenue vs Profit Paradox

Uber Eats emerges as the top-performing channel in terms of revenue across all segments, yet it fails to deliver proportional profit.

This indicates:

- Strong customer demand
- Weak monetization strategy

Strategic Options:

1. Reprice SKUs to absorb commission impact
2. Introduce cross-channel migration tactics, such as:
    - Coupons inside Uber Eats orders
    - Incentives to shift customers to Self-Delivery or In-Store

This allows the company to retain customers while reducing commission dependency.

In-Store Growth Lever

The Full-Service segment is already showing strong In-Store performance, indicating:

- Good brand recall
- Strong offline demand

Recommendation:

- Increase targeted ad campaigns
- Focus on local marketing and footfall conversion strategies

Secondary Objectives & Strategic Support

1. Data-Driven Channel Prioritization

- Primary focus: In-Store and Self-Delivery (profit drivers)
- Secondary role: Uber Eats and DoorDash (customer acquisition channels)

Aggregators should be leveraged for:

- Increasing visibility
- Introducing products to new customers
- Driving trial, not dependency

2. Financial Case for Aggregator Negotiation

- 61% revenue contribution vs ~6% profit contribution
- This imbalance provides strong financial justification for renegotiation

Without intervention, the business risks becoming volume-heavy but margin-poor.

3. Self-Delivery ROI Insights

- Moderate ROI category (<4) contributes 78.12% of total profit

High ROI category contributes 13.41%, with premium pricing but lower volume

Key Insight:

- Profitability is driven by volume-efficient moderate ROI customers, not just high-margin premium customers

Critical Risk Indicators & Red Flags

Commission Burden

- High commission brackets (≥30%) dominate cost structure
- Immediate need to reduce commission rates by at least 10%

Negotiation Impact Scenarios

- 5% commission reduction → Profit share jumps from 6% to 27%
- 3% reduction → Profit increases to ~20%

Even small reductions have exponential impact on profitability

Strong Negotiation Leverage

- Daily orders:
  - **Uber Eats : 26,679**
  - **DoorDash : 14,538**

Implication:

- The company is a high-value partner
- Aggregators are unlikely to risk losing such volume

Strong position to negotiate better terms

Negative Growth Correlation

- Correlation between orders and profit: -0.16 (aggregators)

Meaning :

- Increasing orders → increasing losses

Current model is not scalable in its present form

Self-Delivery Growth & Optimization Strategy

A clear opportunity exists to increase profitability through AOV (Average Order Value) expansion:

Recommended tactics:

- Offer discounts on higher cart values
- Introduce free delivery thresholds
- Create bundled product offerings

Higher AOV directly improves unit economics and net profit

Channel Synergy Insight

A positive relationship exists between:

- In-Store revenue
- Self-Delivery performance

Interpretation :

- Strong brand presence drives both online and offline demand
- These channels reinforce each other, forming a stable ecosystem

Streamlit Web Application (Decision Simulation Layer)

To extend beyond visualization, a Streamlit-based interactive web application was developed to enable real-time scenario analysis and business experimentation.

Core Modules

1. Channel-Wise Profit Comparison

- Compare profitability across all channels.
- Highlights true profit drivers vs revenue drivers

2. Margin Waterfall Charts

- Visual breakdown of:
  - Revenue
  - Commission deductions
  - Delivery costs
  - Final net profit

Clearly shows where margin erosion happens

3. Cost Component Breakdown Panels

- Detailed view of:
  - Commission cost
  - Delivery cost
  - Operational cost

Helps identify optimization opportunities

4. Cuisine & Segment Profitability Heatmaps

- Profitability across:
  - Different cuisines
  - Restaurant segments

Enables targeted strategy per category

Data Integrity Note

- No significant outliers detected across datasets
- Revenue, order volume, and cost distributions are consistent
- Insights derived are reliable and actionable

Final Strategic Direction

To ensure long-term profitability and sustainability, the company should:

- Double down on high-margin channels:
  - In-Store
  - Self-Delivery
- Reposition aggregators as strategic tools:
  - Customer acquisition platforms, not profit drivers

a. Execute immediate action plan:Negotiate commission reductions (target 5–10%)
b. Reprice SKUs on aggregator platforms
c. Shift customers to owned channels
d. Fix negative margins in Full-Service segment
e. Scale Ghost Kitchen operations aggressively
f. Increase Self-Delivery AOV through pricing and offers

Stream lit Dashboard

Orders	Revenue	Net Profit	Profit Margin	Commission	Delivery Costs	Distance (in KM)
436,135	16,789,381	151,122	0.90%	5,041,467	0	0

Select Channel

- In-Store
- DoorDash
- Uber Eats
- Self Delivery

Cuisine Types

- [x] (All)
- [x] Burgers
- [x] Chicken Dishes
- [x] Chinese
- [x] Indian
- [x] Japanese
- [x] Kebabs/Mediterranean
- [x] Pizza
- [x] Thai

Segment

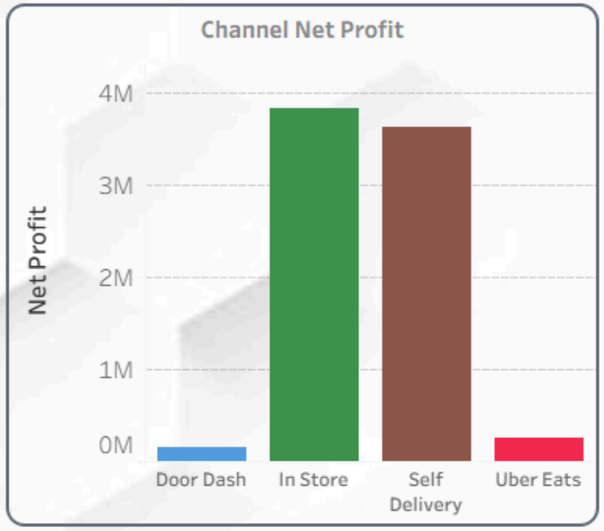
- [x] (All)
- [x] Cafe
- [x] Full-service
- [x] Ghost Kitchen
- [x] QSR

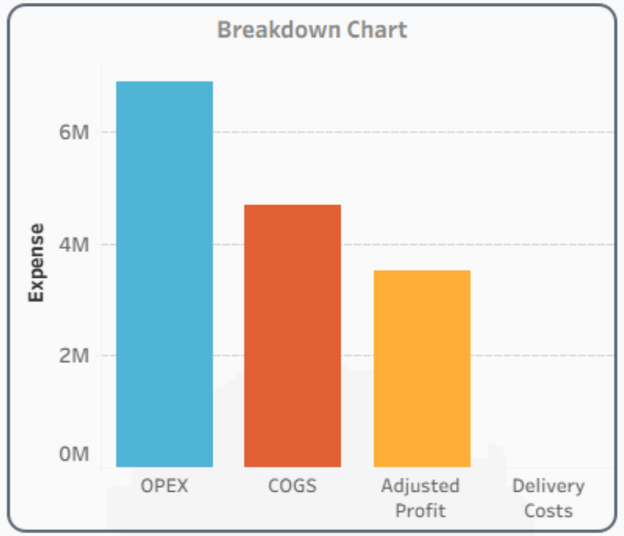
Commission Slider

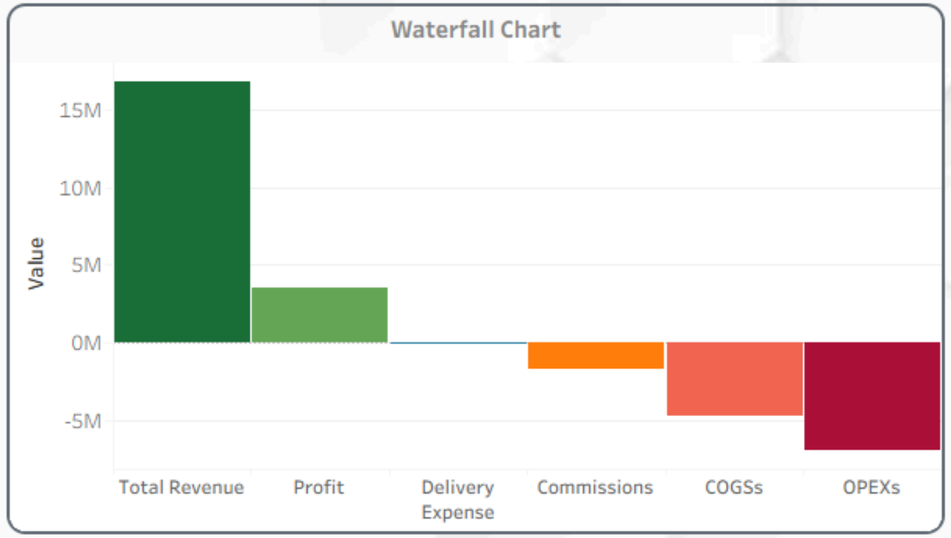
0.1

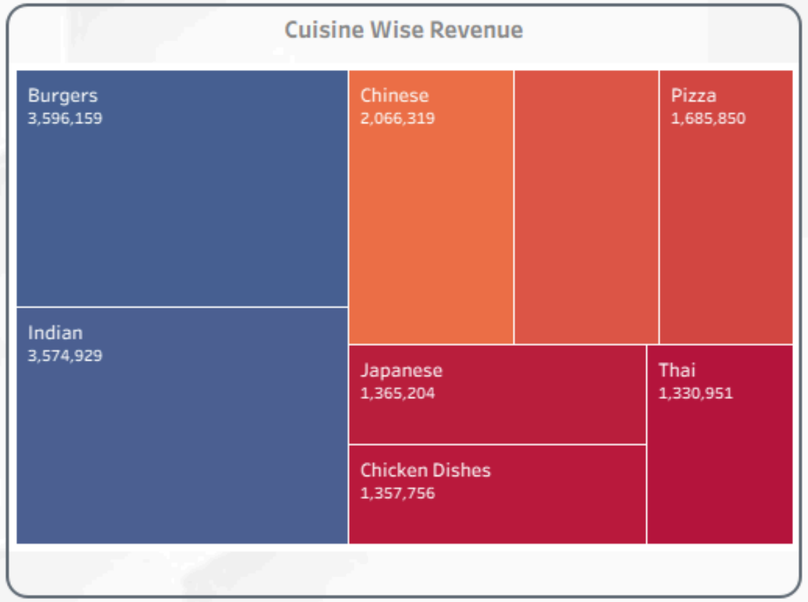
Delivery Cost Slider

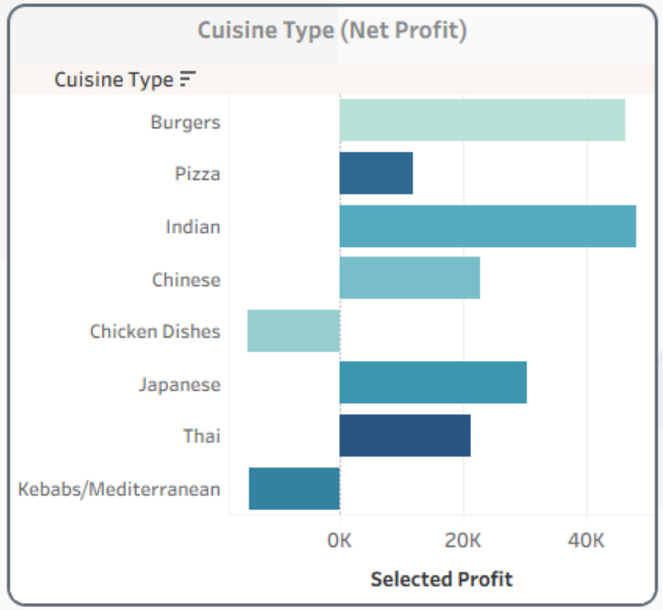
0.9

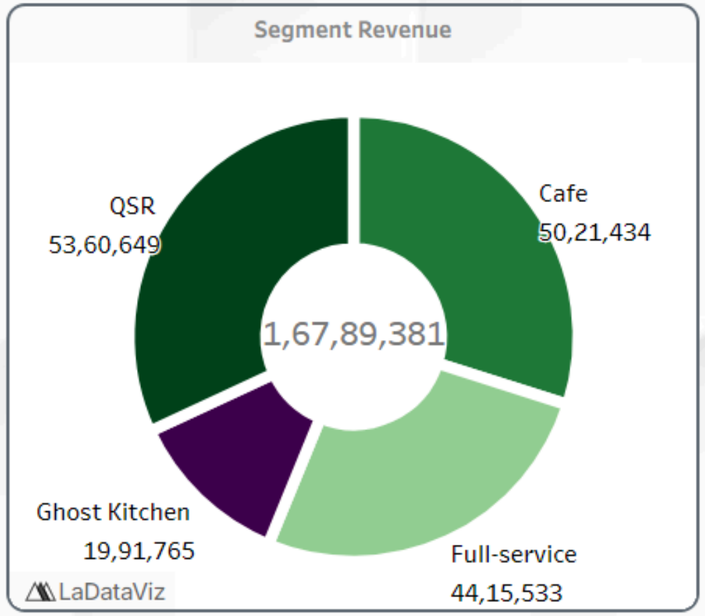












Conclusion

The overall analysis highlights a clear imbalance between revenue generation and profitability, where aggregator platforms like Uber Eats and DoorDash contribute significantly to top-line growth but fail to translate that into sustainable profits. Despite generating more than 60% of total revenue, their contribution to net profit remains minimal due to high commission structures and delivery-related costs, making them inefficient from a margin perspective.

On the other hand,

Channel	Net Profit	Profit Contribution
In Store	3828417.36	48.67
Uber Eats	258488.08	3.29
Door Dash	151122.25	1.92
Self Delivery	3628278.44	46.12

In-Store and Self-Delivery channels emerge as the strongest pillars of profitability, collectively contributing nearly 95% of total profit. These channels provide better control over pricing, customer experience, and cost structure, making them essential for long-term business sustainability.

At a segment level, while Ghost Kitchens demonstrate strong and scalable profitability, Full-Service restaurants require immediate attention due to negative margins on aggregator platforms, and Café/QSR segments need pricing optimization to improve margin realization. Additionally, the negative correlation between order volume and profitability in aggregator channels indicates that continuing the current growth strategy without intervention will further increase losses over time.

From a strategic standpoint, the business should shift its focus from revenue-heavy to profit-driven channels, while repositioning aggregators as customer acquisition platforms rather than core revenue drivers. Immediate actions such as commission renegotiation, pricing adjustments, and customer migration to owned channels can significantly improve financial outcomes.

Furthermore, the integration of a Tableau dashboard for insights and a Stream lit-based application for real-time scenario analysis transforms this project into a decision intelligence system, enabling stakeholders to make data-driven, scenario-based strategic decisions.

In conclusion, by optimizing channel strategy, improving cost structures, and leveraging data-driven tools, the company can move from a high-volume, low-margin model to a balanced, sustainable, and profit-focused growth trajectory.

Recommendations

Scale High ROI Restaurants :

- Expand SKUs focused on high-volume bestsellers and optimize menu mix to maximize revenue and margins.

Fix Loss-Making Restaurants :

- Conduct detailed audits and on-ground reviews to identify pricing gaps, cost leakages, and operational issues.

Mentor Low ROI Restaurants :

- Provide targeted guidance on pricing, cost control, and improving both offline (In-Store) and online performance.

Optimize Moderate ROI Restaurants :

- Reduce delivery costs by positioning closer to high-demand areas and improving logistics efficiency.

Aggressive Commission Negotiation :

Form a unified/vendor coalition approach to negotiate lower commission rates with Uber Eats and DoorDash, leveraging high order volumes to push for 3–10% reduction and improve overall profitability.

Flexible Data Availability :

Any additional data required for deeper analysis can be provided upon request. This includes High ROI restaurant lists, channel-wise and segment-wise net profit breakdowns, restaurant-level performance insights, and revised profitability scenarios based on adjusted commission rates, enabling more granular and customized decision-making.

Thank You